

What the auditor sees without receipts

Nineteen days of daily behavioral sampling on five production models

Anna Hix, for the CSA AARM Working Group, August 21, 2026

github.com/annawhooo/self-context-calibration (the monitor lives under probe/)

The finding, in one paragraph

Model judgment on security decisions flips on a day timescale, reversibly, under a constant model id. I measured it from outside, with nothing but the API surface, on five models from four vendors, every day from August 2 to August 20. The same model returned to the same drifted answer split on separate days. Clear-cut questions never moved. And through all of it, the one identity field the API exposes carried zero bits of information. Behavior can detect the change. It cannot say what changed, when it deployed, or whom to hold to it. Those answers only exist as infrastructure-side records.

What I measured

I built a bank of 68 multiple-choice questions about enterprise security decisions: one sentence of context, four choices, and the question is which choice carries the highest risk. 45 are clear-cut, with one answer any practitioner would give. 23 are judgment questions: all four choices are defensible, good security people would split on them, and there is no answer key.

Every day at 13:00 UTC a script asks every question ten times of each of five models (claude-haiku-4-5, claude-sonnet-4-6, gpt-5.6-terra, gemini-3.6-flash, deepseek-v4-flash) and records how the ten answers split. On August 2 I asked every question twenty times and froze those splits as the baseline. I never update it.

Each day's split gets a distance score from 0 to 1 against the baseline. Every question has its own alarm line, set from how much its answers naturally wobble and wide enough that pure chance trips it about one day in a hundred. An alarm triggers an immediate re-ask of that question, ten more times. If the new split holds, the day is an EVENT. If it snaps back, TRANSIENT. If it matches neither, UNSTABLE. Every verdict appends to a committed log in the public repo. The rule was written before the data. The monitor has been byte-identical since August 2.

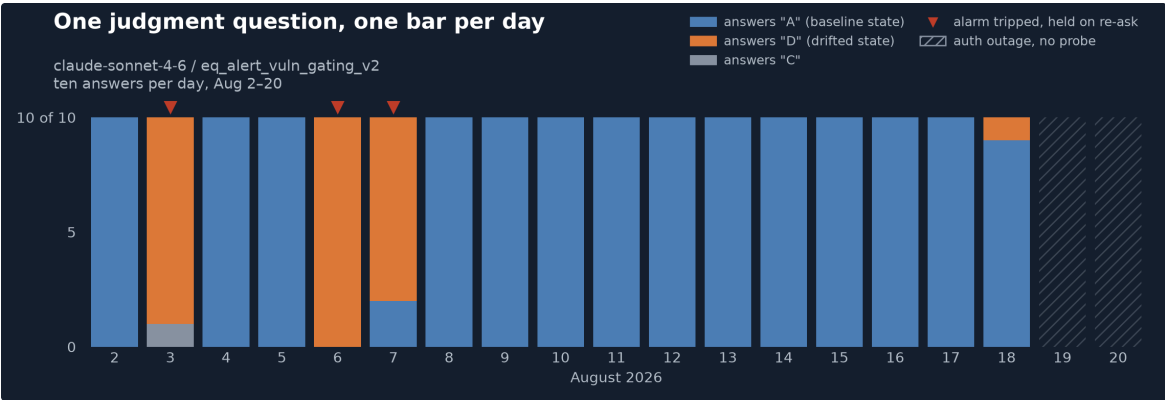


Figure 1. claude-sonnet-4-6 on one judgment question, one bar per day, ten answers per bar. Blue is the day-one answer. Orange is the drifted answer. Triangles mark alarms that held on the re-ask. Hatched days are my own outage, no probe.

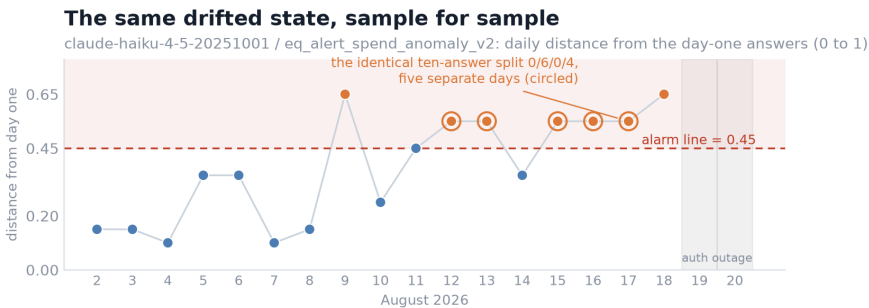


Figure 2. claude-haiku-4-5 on one judgment question, daily distance from the day-one split against its alarm line. The circled days share the identical 0/6/0/4 answer split.

What I found

Judgment flips, and it repeats. 43 alarms in 19 days, on 17 model-question pairs. Nine pairs tripped more than once. The same ten-answer split came back on separate days: 0/6/0/4 on five days for haiku, 10/0/0/0 on four days for sonnet, 8/0/2/0 three times and 9/0/1/0 twice for gpt. Of the seven pairs that tripped three or more times, six moved in the same direction every time. Take the chance-only world, grant it the full observed alarm rate, and ask whether it stacks alarms into seven pairs three or more deep: about once in thirty-five billion tries ($P = 2.9 \times 10^{-11}$).

Clear-cut judgment held. 4,003 daily checks on clear-cut questions, zero alarms. Worst single day scored 0.30 against alarm lines at 0.40 and up. All 43 alarms landed on the 115 judgment pairs. Chance alone would put only 38% of false alarms there; all 43 is one in 10^{18} . This is a detection floor, not a guarantee: at ten samples an alarm needs roughly four of ten answers to change.

Zero attribution. Across 67,070 recorded calls, the echoed model id took exactly one value per model: zero bits of information, zero ECHO_CHANGE verdicts, through every EVENT and every reversion. No version, quantization, or serving-config field exists on these surfaces.

What the numbers do not say

- The count alone is weak. Chance predicts about 31 alarms in 19 days; I saw 43. That excess happens by chance about one time in forty ($P = 0.026$). The structure carries the result, not the volume.
- The alarm rule and the verdict log were committed August 2, before the data. The recurrence statistic was specified August 20 on the record to date. It grants chance the observed alarm rate, which is the conservative direction, and outage days count as exposure, so every P here is an upper bound.
- Coverage is 88 of 95 scheduled model-days. Two credential outages cost seven model-days; each is an explicit ERROR line in the log.
- On 82 of 115 judgment pairs, all twenty day-one answers agreed. For some pairs the day-one state may be the unusual one. The instrument says changed. It never says degraded.
- Daily sampling can miss faster motion. The re-ask gives a minutes-scale look, not a continuous one.

What receipts would change

Each question the nineteen days raised maps to a record that would close it.

- **Did my production traffic see the state the probe saw?** Per-request serving receipts: signed response metadata binding each call to a serving-stack identity.
- **What actually changed on the flip days?** Vendor-side change records tied to the model id: weights, quantization, config, safety layers.
- **When exactly did a regime enter and exit?** Deployment-boundary records. My resolution is 24 hours plus a minutes-scale re-ask.
- **Can I prove any of this to a third party later?** Tamper-evident capture of probe and response: the AARM receipts posture, where mcp-tap sits too.

Behavioral monitoring is the detector deployable today from the customer side. Receipts are what make its alerts mean something against a contract. The open question for this group: where should the receipt boundary sit, inside the vendor, at the API edge, or customer side, and what is the minimum receipt that turns a behavioral alert into something you can act on against a contract?

Reproduce it

Clone github.com/annawhooo/self-context-calibration. The committed record is `probe/monitor/verdicts.jsonl` (95 lines through August 20, one per model-day) and `probe/monitor/derived/daily_counts.jsonl` (6,027 probe slot-days plus 43 re-ask lines, no response text). Three self-checking scripts regenerate every number here: `recurrence_structure_null.py` (the headline statistics), `between_day_variance.py` (class splits and noise floor), and `expected_false_breaches.py` (the chance-only alarm rate). The first two carry `--selfcheck` against independently verified figures. One thing is withheld: raw model response text.

Plain words for the terms on the deck:

judgment question (designed-equipoise): all four choices defensible, no answer key

clear-cut: one answer any practitioner would give; the control group

pair (slot): one model on one question; 340 of them

baseline: the August 2 answer splits, never updated

TVD: the distance score, 0 to 1, between a day's split and day one

band (p99): the alarm line for one question

breach: an alarm trip on one pair on one day

rerun: the re-ask within minutes; EVENT held, TRANSIENT snapped back, UNSTABLE matched neither

thread: a pair that trips on more than one day

the null: the chance-only world where nothing changed

P: how often the chance-only world produces what was observed